Autism: when to test
Indian trio data: exome found a cause in about 3 in 10 children

For sales and healthcare teams. Facts summarised in our own words from published sources (links below). Treatment decisions are always the doctor's.

Indian study: Sheth F, Shah J, Jain D, et al. *Comparative yield of molecular diagnostic algorithms for autism spectrum disorder diagnosis in India: evidence supporting whole exome sequencing as first tier test.* BMC Neurology 2023;23:292. doi:10.1186/s12883-023-03341-0. FRIGE's Institute of Human Genetics, Ahmedabad. Open access (CC BY 4.0).

Who was tested

- 101 children with idiopathic autism (DSM-5) and their parents (trios); children with prominent syndromic features excluded
- Every child had karyotype, Fragile X (boys), microarray and exome

What was found

- Karyotype 0%; Fragile X 0%; microarray 2.9% (3/101); **exome 29.7% (30/101)**, which also detected all the microarray findings
- **With seizures 53.6%** (15/28) vs without 20.5% (15/73)
- Most frequent gene: **MECP2 (Rett syndrome)**, 6 cases
- About two-thirds of causes were de novo; VUS in 20.8%

Second Indian study (Mumbai)

Trio exome in 137 children with autism: a pathogenic or likely pathogenic variant in **16.1%**; **35%** in children with syndromic features vs **8.2%** with autism alone; 91.3% of significant variants were de novo (Bajaj et al., J Hum Genet 2025). The yield depends on who is tested: children with seizures, dysmorphic features or intellectual disability are the most likely to get an answer.

Key points for doctors

- Autism is diagnosed clinically; a genetic test looks for the underlying cause.
- Guidelines advise offering genetic testing to every child with autism; microarray and Fragile X are the traditional first tests (AAP 2020). The Ahmedabad authors argue for exome first in India.
- A cause informs recurrence risk and, for some genes, seizure treatment or monitoring.
- Prenatal testing in a later pregnancy: disease only; fetal sex never disclosed (PCPNDT).

EPISODE 12 · SALES FIELD SHEET

Autism: how to approach doctors and centres

Topic	What to do or say
Doctors and centres	Paediatricians, paediatric neurologists, developmental paediatricians, child psychiatrists and psychologists; autism therapy, speech and occupational therapy centres; special schools.
Which children	Seizures; dysmorphic features or birth defects; intellectual disability as well; girls who lose skills (think Rett syndrome).
Opening question	"Doctor, do you test autistic children who also have seizures?"
Facts to quote	Ahmedabad: exome 29.7% vs microarray 2.9%; with seizures 53.6%. Mumbai: syndromic 35% vs autism alone 8.2%.
"Therapy is the real treatment"	"Yes, therapy continues. A cause adds recurrence risk and can change seizure care."
"Reports come back uncertain"	"Trio testing gives clearer reports, and no major decisions are taken on a VUS."
Sample	EDTA blood; parents' samples for trio; counselling and written consent first.
Never say	That the test diagnoses or cures autism, or guarantees an answer. Quote only sourced facts; confirm price and report time before quoting.

References

- Sheth F et al. BMC Neurology 2023;23:292. doi.org/10.1186/s12883-023-03341-0
- Bajaj S et al. Trio exome sequencing in 137 Indian children with autism spectrum disorder. J Hum Genet 2025;70:611-624.
- Hyman SL, Levy SE, Myers SM; AAP Council on Children with Disabilities. Identification, evaluation, and management of children with autism spectrum disorder. Pediatrics 2020;145(1):e20193447.

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